



Figure 6: osquery

Suricata (suricata.io/)

This open source IDS/IPS tool has threat detection features, and integrates with AI for deep analytics of data and digital infrastructure.

Yara ([virustotal.github.io/yara/](https://github.com/VirusTotal/yara))

Yara is focused on malware and classifies assaults associated with specific network signatures. Rule-based mechanisms can

be programmed into this tool so that attacks can be evaluated effectively.

OpenVAS (openvas.org/)

OpenVAS is a multi-featured scanner for identifying different types of vulnerabilities and attacks in the network environment. It can evaluate and prioritise different protocols.

ClamAV (clamav.net/)

ClamAV is a high-performance platform used for the detection and deep evaluation of malware, viruses, Trojans and malicious threats. It can be used for scanning files and signature analytics.

Endpoint security and log monitoring tool

osquery (osquery.io/)

osquery is used for testing, analytics and

endpoint security, and focuses on thread safety and memory leaks. It can be used to query systems to capture critical information, which is then analysed for forensics-based applications and predictive analytics.

These tools and frameworks can be used depending upon the security issue being addressed such as malware, Trojans, vulnerabilities, log files, and so on. **END** 🐧

References

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By: Dr Gaurav Kumar

The author is associated with various academic and research institutes for delivering expert lectures and conducting technical workshops on the latest technologies and tools.

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This information is critical when determining the attack surface and selecting appropriate security measures.

Service version detection: By probing services more deeply, Nmap can often determine the exact version of a service running on an open port. This level of detail is vital for identifying known vulnerabilities associated with specific software versions.

Nmap scripting engine (NSE): The NSE extends Nmap's functionality by allowing users to write scripts that automate a wide range of tasks—from vulnerability detection to compliance checks. With hundreds of community-contributed scripts available, NSE transforms Nmap from a simple scanner into a comprehensive security auditing framework.

Stealth and evasion techniques: Nmap includes features that allow for stealth scanning, which can help avoid detection by intrusion detection systems (IDS) or firewall rules. Techniques such as TCP SYN scanning, fragmenting packets, and using decoys can be particularly useful in both penetration testing and defensive scenarios.

Customisation and automation: Nmap is highly customisable, allowing security professionals to tailor scans to meet specific needs. Combined with automation tools and scripts, Nmap can run regular scans at predetermined

intervals, ensuring that the network's security posture is continuously monitored.

These advanced capabilities underscore why Nmap remains a favourite among security professionals. Its flexibility means that it can be adapted to almost any environment, whether it is a small business network or a complex enterprise system with multiple layers of defence.

In summary, Nmap and port scanning are not merely tools—they are integral components of a comprehensive cybersecurity strategy. By enabling detailed network inventory, effective vulnerability assessment, rigorous compliance auditing, and proactive intrusion detection, they provide the intelligence needed to navigate the complex challenges of modern digital security. Embracing these tools, and understanding their advanced functionalities and ethical implications, is essential for any organisation committed to safeguarding its assets in an increasingly interconnected world. **END** 🐧

By: S.V. Rohith

The author possesses six years of domain experience in RPA, Python, automation, Linux, DevOps, security operations, JavaScript, and has authored over five articles across various domains. Additionally, he is a skilled instructor, guiding over 100 students through his expertise.